6.8 Shapes, maps and graphs: answers

← Unit overview · Practice

For parents: for views and nets, building or folding is the best way to check. If he disagrees with an answer here, test it together. Being able to convince you with a model is excellent maths.

Set 1 · Warm-up

#	Answer	How
1	3	
2	6	Top, bottom, front, back, left, right.
3	B2	Column B, row 2.
4	The library	
5	E1	
6	A5	
7	The pool	
8	4 squares	A5 → B5 → C5 → D5 → E5.
9	4	
10	The park	

Set 2 · Scale

#	Answer	How
1	3 km	3 × 1 km.
2	7 km	7 × 1 km.
3	500 m	5 × 100 m.
4	8 km	4 × 2 km.
5	5 cm	10 km : 2 km = 5.
6	2 000 m (2 km)	4 squares × 500 m.
7	2 000 m (2 km)	E1 → E5 is 4 squares.
8	4 squares, 2 000 m (2 km)	A5 → A2 is 3 squares, then 1 square to B2. 4 × 500 m.
9	2 m	2 × 1 m.
10	1 km	

Set 3 · Views and nets

#	Answer	How
1	8 blocks	3 + 1 + 2 + 1 + 1.
2	3 blocks	Back left.
3	A	From the front you see the tallest tower in each column: left 3, middle 1, right 2.

#	Answer	How
4	Yes	The cross.
5	No	A 2 × 3 rectangle: when you fold it, squares overlap and one side stays open.
6	Yes	A row of 4 with one square above and one below (on different ends) always works.
7	No	Five in a row is too many: the ends overlap when you fold.
8	6	One for each face.

Set 4 · Graphs

#	Answer	How
1	60 bikes	6 × 10.
2	55 bikes	5 × 10 + 5.
3	Wednesday	30 bikes. Many Dutch primary schools finish at lunchtime on Wednesday, so perhaps fewer children cycle, or some don't come back after lunch. Any sensible reason is fine.
4	40 more	70 − 30.
5	265 bikes	60 + 55 + 30 + 50 + 70.
6	16 °C	
7	15:00	18 °C, the highest point.
8	10 degrees	18 − 8.
9	18:00	
10	Between 9:00 and 12:00	It rose 6 degrees (10 → 16). Between 6:00 and 9:00 it rose 2, and between 12:00 and 15:00 it rose 2. The steepest part of the line.

Talk about it: the difference is only 3 ice creams (55 − 52). Starting the axis at 50 makes it look huge. The shop might want Sunday to look like a big success. This is exactly the kind of critical reading the new national goals ask for.

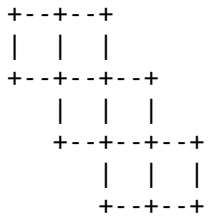
Set 5 · The ratio table

#	Answer	How
1	12, 24, 48	Double each time, top and bottom.
2	500, 125, 750	8 people: double. 2 people: half. 12 people: 3 × 4 people, so 3 × 250.
3	24, 40, 80	3 × 8, 5 × 8, 10 × 8.
4	15, 35	3 × 5, 7 × 5.
5	2,40 and 0,40	6 apples: double. 1 apple: €1,20 : 3 = 40 cents.
6	30 km in 2 hours; 7,5 km in half an hour	Double 15; half of 15.
7	20, 30	× 10.

Talk about it: from 4 to 2 people you halve both rows. From 4 to 12 you multiply both rows by 3. (Or: go via 8 and add 4 people's worth.)

Puzzle corner

There are **11** different cube nets. Six of them are "a row of 4 with one square above and one below" in different positions; the other five are different shapes, including the "staircase":



Finding one new net is a real success; three is excellent. Proving there are exactly 11 is genuinely hard (adults find it hard too). The key ideas are: never four squares in a 2 × 2 block, and never five in a row. If he gets organised (for example "nets with a row of 4", then "nets with a longest row of 3", then "longest row of 2"), that's mathematical thinking at its best.